

EE / CprE / SE 491 — sdmay27-11

Weekly Report 1:

MIMO FMCW Radar

14th – 21st September 2026

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§1 Summary

The team finished the system block diagram and selected the main IC for each block. The DAC and VCO boards are complete through layout, and the TX up-converter schematic is done with layout in progress. Waveguide and transition design started, with the Vivaldi being evaluated as the microstrip-to-waveguide transition; it is not yet matched across 24 to 40 GHz. Signal processing defined the IF capture chain, and the FPGA sub-team set up an Opal Kelly board and planned the SPI controller.

§1.1 Team and Roles

Team Member	Role
Flatrim Dreshaj	Team Leader
Justin Adrowski	FPGA Programming
Durotimi Johnson	TX/RX Layout Design
Ryland Feist	VCO / PLL / Synthesizer
Elliott Bourgeois	Signal Processing
Brett Brunner	Antenna Design
Aidan O'Grady	FPGA / MCU Programming

Table 1. Team members and roles

§1.2 Status at a Glance

Work Item	Owner	Status
System block diagram and IC selection	Flatrim Dreshaj	Complete
DAC fast-sweep board	Flatrim Dreshaj	Schematic and layout complete
Ethernet SPI board	Flatrim Dreshaj	Schematic in progress
VCO board	Ryland Feist	Schematic and layout complete
TX up-converter board	Durotimi Johnson	Schematic complete, layout in progress
Waveguide and transition design	Flatrim Dreshaj	In progress
Antenna Design	Brett Brunner	In progress
IF capture chain	Elliott Bourgeois	Architecture defined
FPGA SPI controller	Justin Adrowski, Aidan O'Grady	Research done, not yet programming

Table 2. Status of each work item at the end of the week

§2 System-Level Design

The radar is a homodyne FMCW design. A DAC ramp sweeps the VCO to produce a linear chirp, and the received echo is mixed with that same chirp. The echo delay appears as a beat frequency, which gives the range.

§2.1 Block Diagram

Figure 1 shows the signal path. The FPGA drives the DAC, which tunes the VCO. The TX chirp passes through the up-converter, a microstrip-to-waveguide transition and a Ka-band waveguide to the antenna, and the RX path mirrors this into the down-converter. The ADCs digitize the IF signal for the FPGA, which also configures the converters.

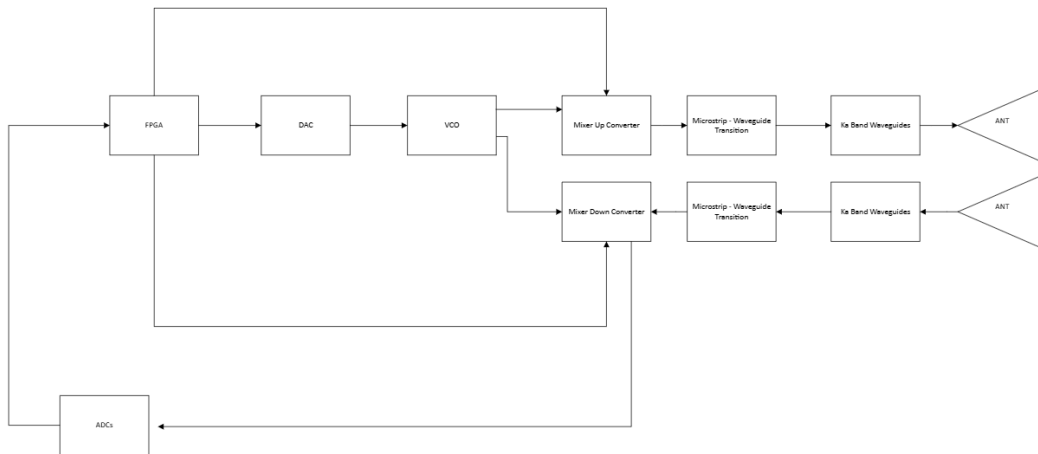


Figure 1. System block diagram of the MIMO FMCW radar

Range resolution depends on the swept bandwidth B :

$$\Delta R = c / (2B) \quad (1)$$

Sweeping the full 24 to 40 GHz band gives $B = 16$ GHz and $\Delta R \approx 9.4$ mm, which is why §4 targets the whole band.

§2.2 IC Selection

Table 3 lists the main parts.

Block	Part	Purpose
Sweep DAC	AD3551R	Fast-sweep DAC that generates the VCO tuning ramp

Block	Part	Purpose
DAC output buffer	AD8065	Op-amp that buffers the DAC output
VCO	HMC587LC4B	Wideband VCO that produces the chirp
Mixers	ADAR2001/2004	Transmit/Receive converter with four differential RF outputs/Inputs
Logic translation	TXB0101	Shifts 3.3 V SPI and sequencer logic to 1.8 V
TX regulators	REG1117-1.8, REG1117-2.5	1.8 V and 2.5 V rails on the TX board
FPGA platform	Opal Kelly development board	FPGA development with attached ADC modules

Table 3. Main ICs selected for each functional block

§3 PCB Design

The DAC and VCO boards are finished through layout, the TX up-converter schematic is done with layout underway, and the Ethernet SPI board schematic is in progress.

§3.1 DAC Fast-Sweep Board

The DAC board is a hierarchical KiCad design. Its top sheet also holds research, assembly and testing, and results pages, so bring-up data stays with the schematic (Figure 2).

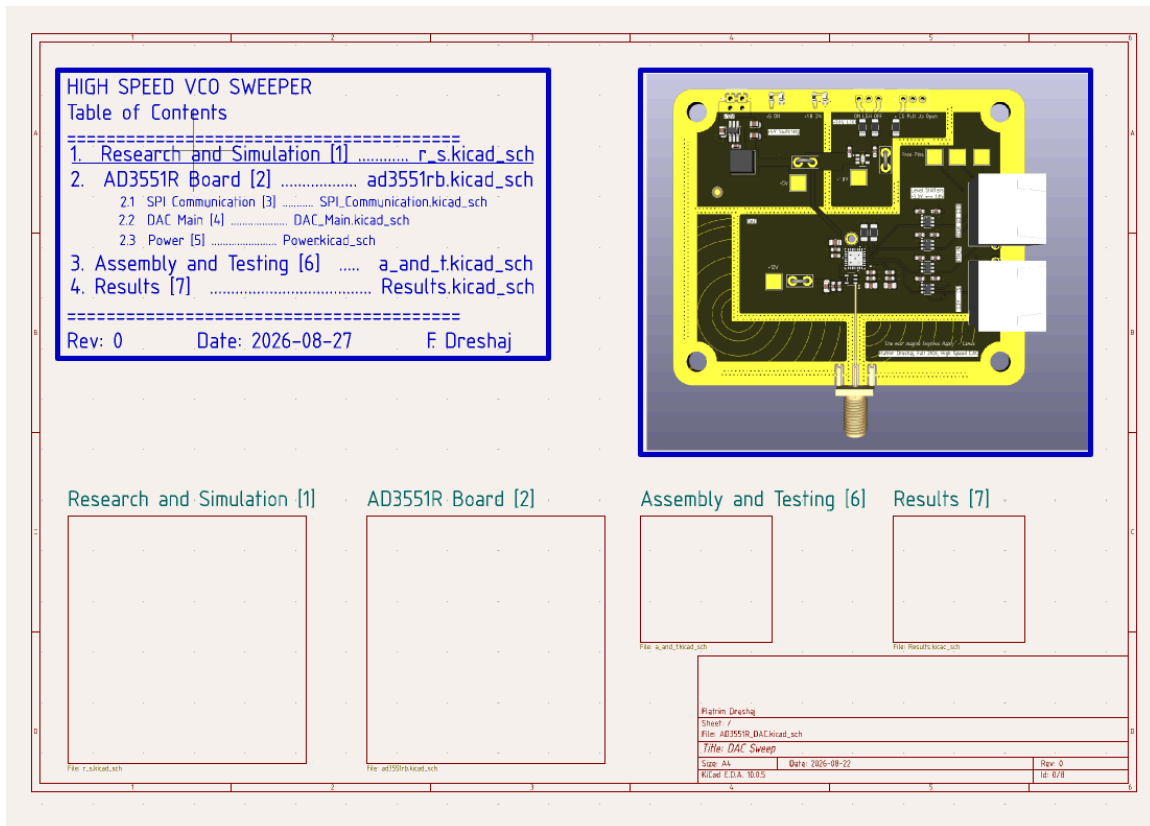


Figure 2. Top sheet of the DAC board project, with its table of contents and a 3D preview

SPI arrives over two shielded RJ45 connectors and power through a screw terminal (Figure 3). The SPI lines pass through level shifters to the DAC, and the DAC output leaves on an SMA connector.

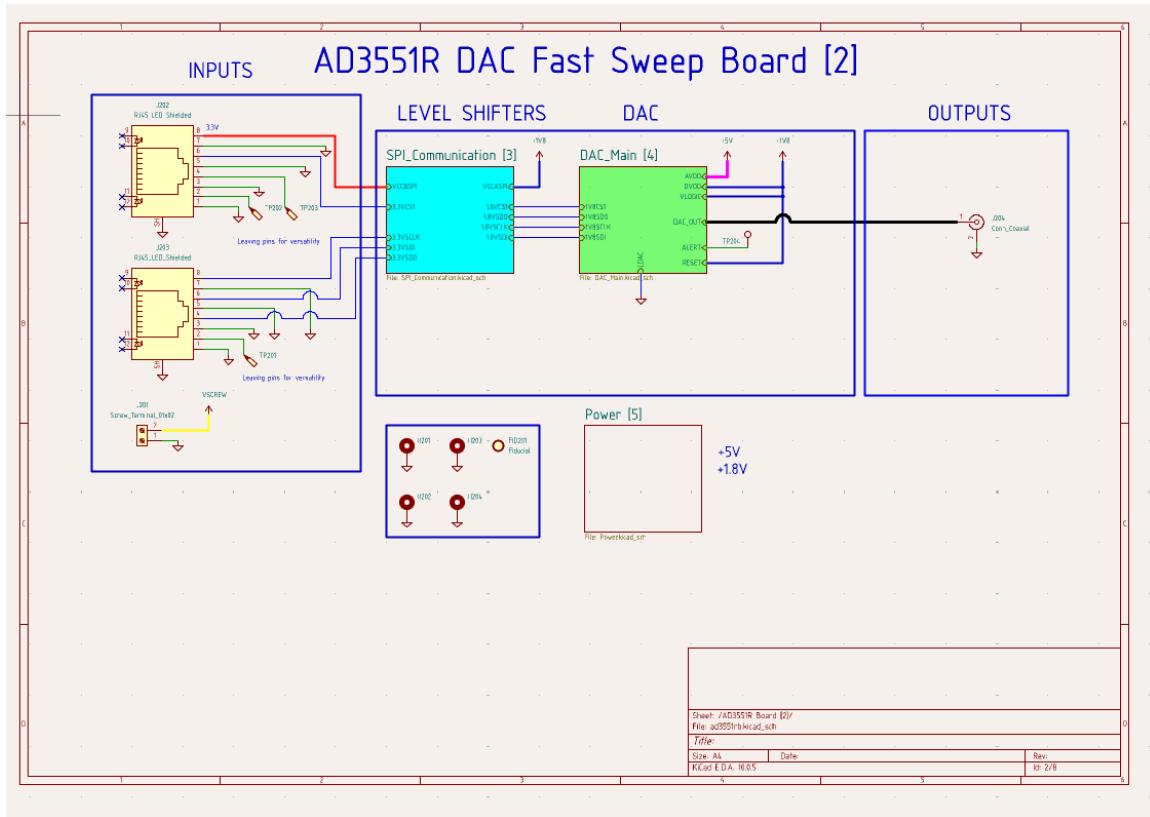


Figure 3. AD3551R board sheet: RJ45 inputs, level shifters, DAC and SMA output

Each SPI line has its own TXB0101 translator (3.3 V to 1.8 V) with 100 nF decoupling on both rails. A switch lets the shifters be disabled during bring-up (Figure 4).

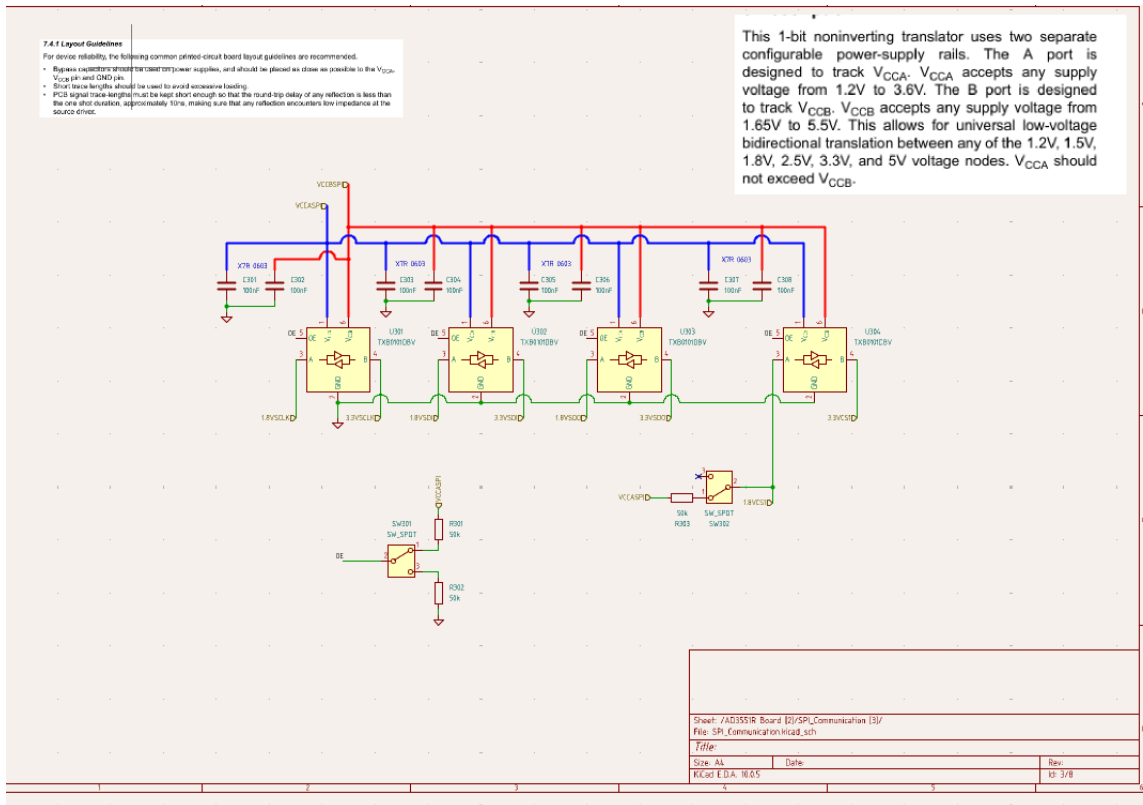


Figure 4. SPI communication sheet with TXB0101 level shifters and the translator datasheet notes

The AD3551R output is buffered by a transimpedance AD8065 op-amp (Figure 5). Layout followed the notes on this sheet: extended QFN pads, a short and thin I_OUT trace, and constant-impedance routing.

Power comes from a 5 V, 2 A switching regulator followed by a 1.8 V, 1 A LDO (Figure 6). The AD3551R (about 61 mA) and AD8065 (about 14 mA) are well within these ratings.

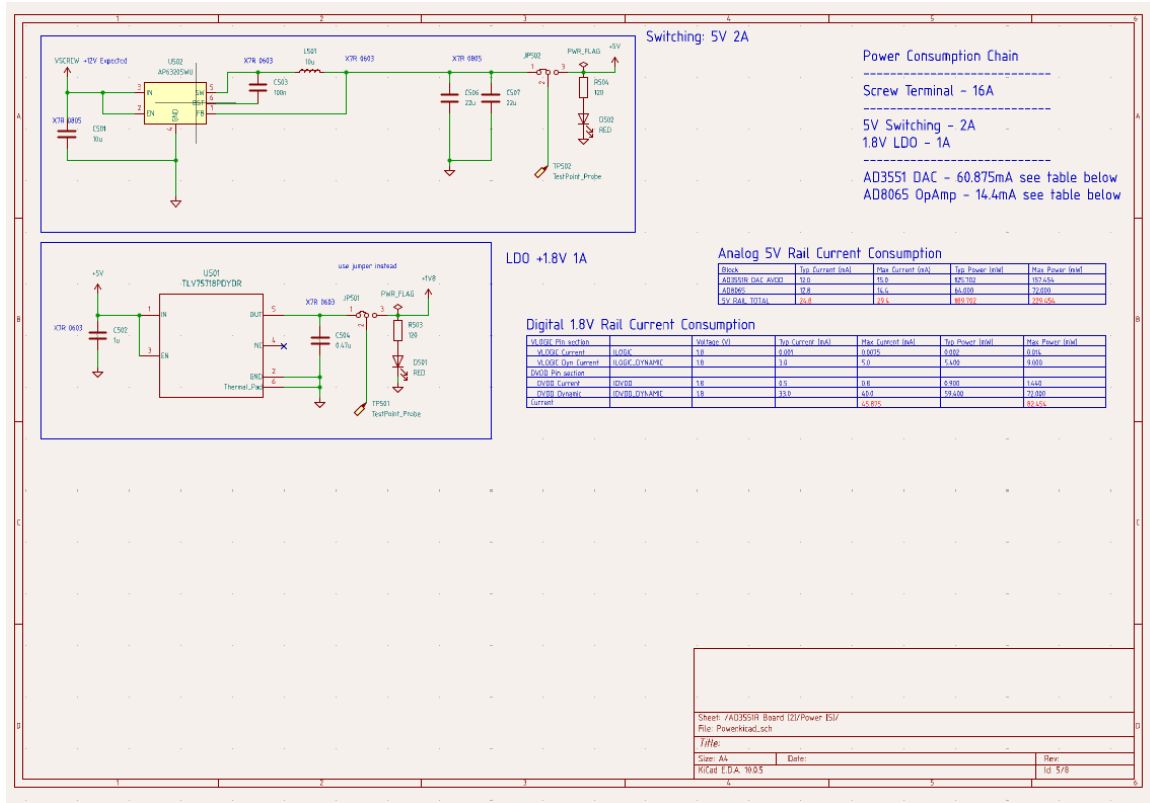


Figure 6. Power sheet with the 5 V switching and 1.8 V LDO stages and the rail current budget

Figure 7 shows the finished layout. Jumpers let each supply rail and the level shifters be enabled one at a time during bring-up.

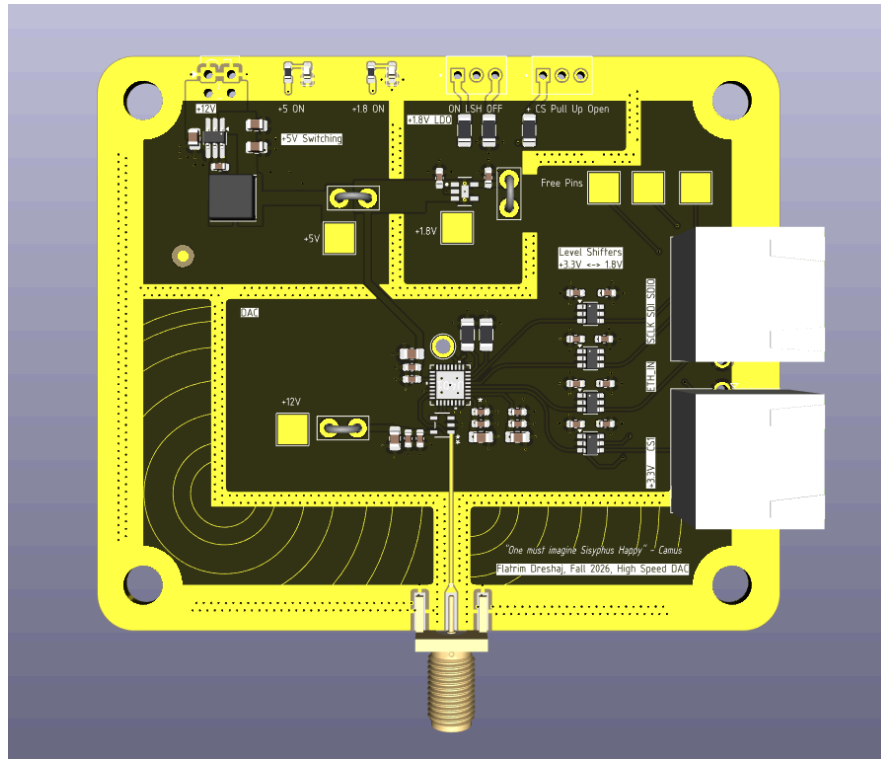


Figure 7. DAC board layout, 3D view

§3.2 Ethernet SPI Board

The schematic is in progress. This board breaks the FPGA's SPI lines out to the RJ45 cables used by the DAC and TX boards.

§3.3 VCO Board

The VCO board (Figure 8) has screw terminals for power and tuning, the HMC587LC4B, decoupling capacitors, and a resistive splitter feeding three SMA outputs.

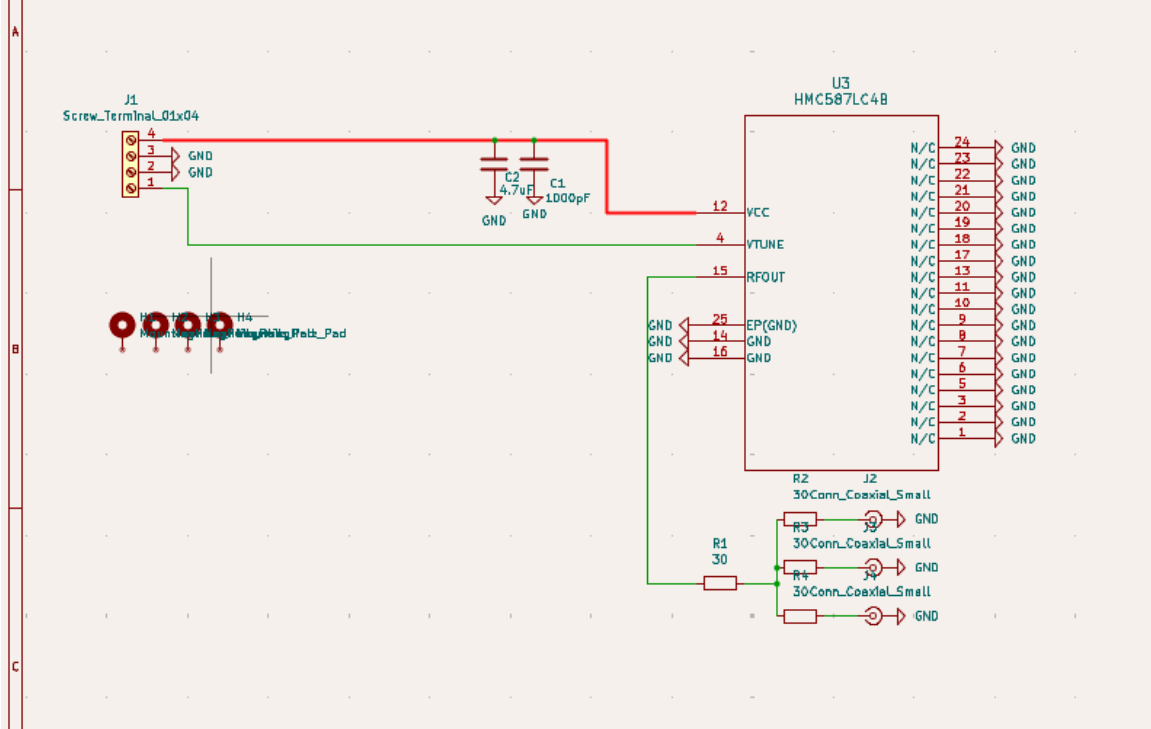


Figure 8. VCO board schematic with the HMC587LC4B and a three-way resistive splitter

The splitter resistors follow the matched star-splitter formula for N ports:

$$R = Z_0 (N - 1) / (N + 1) \quad (2)$$

With $N = 4$ and $Z_0 = 50 \, \Omega$ this gives the $30 \, \Omega$ used. Each output is about 9.5 dB below the input, which sets the drive level at the up-converter.

In the layout (Figure 9), the $50 \, \Omega$ lines to the SMA connectors are via-fenced and have the solder mask removed.

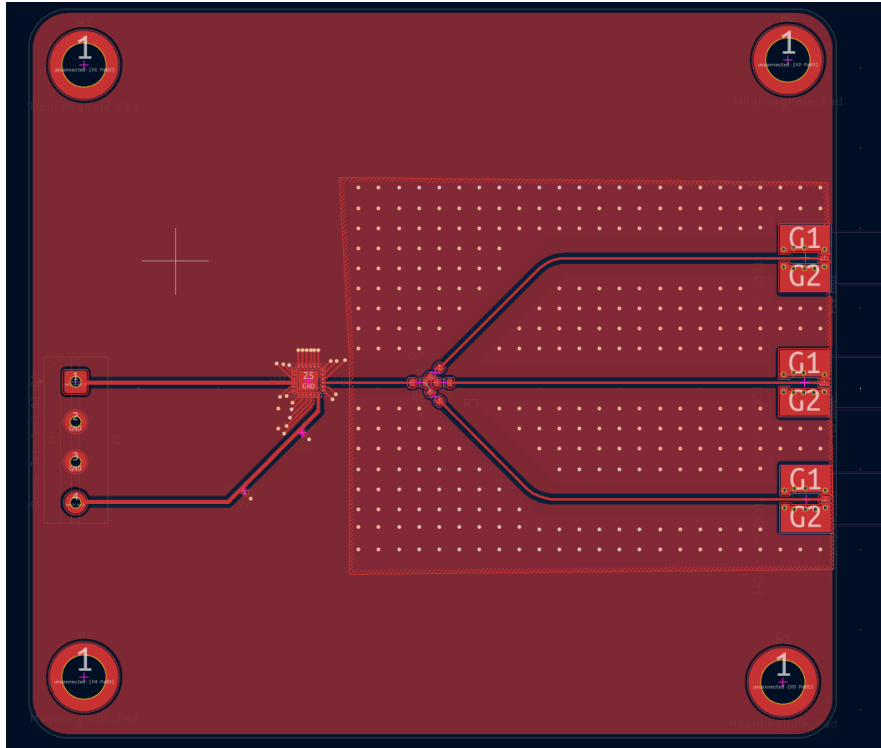


Figure 9. VCO board layout with via-fenced 50 Ω lines to the SMA outputs

§3.4 TX Up-Converter Board

The TX board uses the ADAR2001 (Figure 10). The schematic is complete but may change, with sub-sheets for power, SPI and sequencer level shifting. Each supply pin has a 10 nF and 100 pF decoupling pair at the pin.

The board has coaxial power and RF inputs, sequencer headers, a 120-pin SDP connector, and RJ45 jacks for SPI and sequencer control (Figure 11).

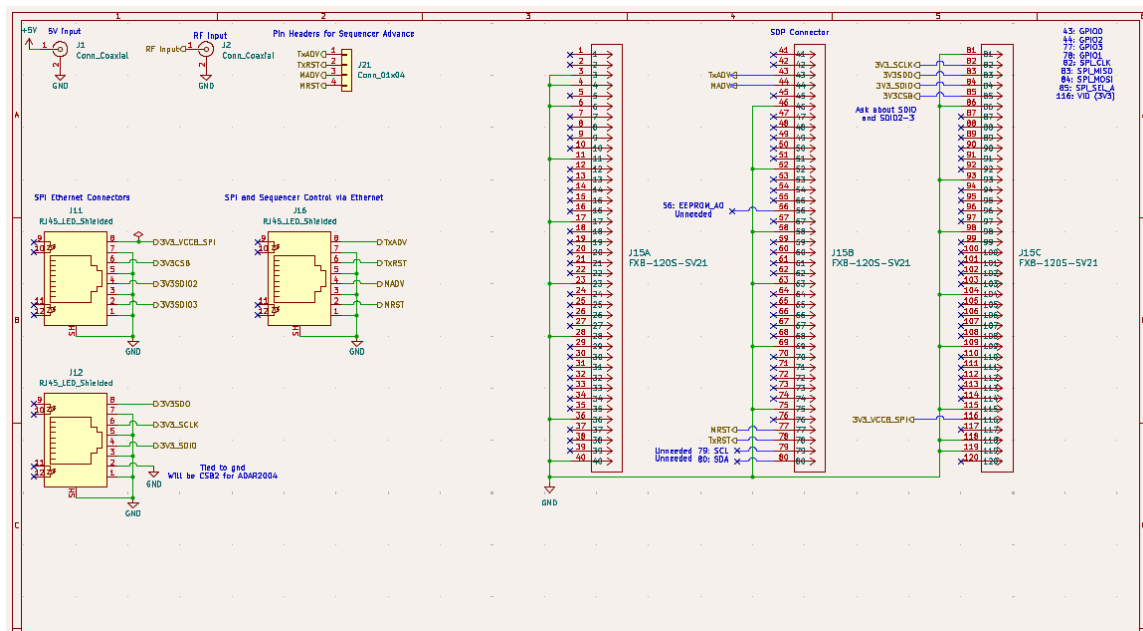


Figure 11. TX board connectors: RF and power inputs, SDP connector and RJ45 control jacks

The 1.8 V and 2.5 V rails use REG1117 regulators built from the datasheet reference circuit (Figure 12).

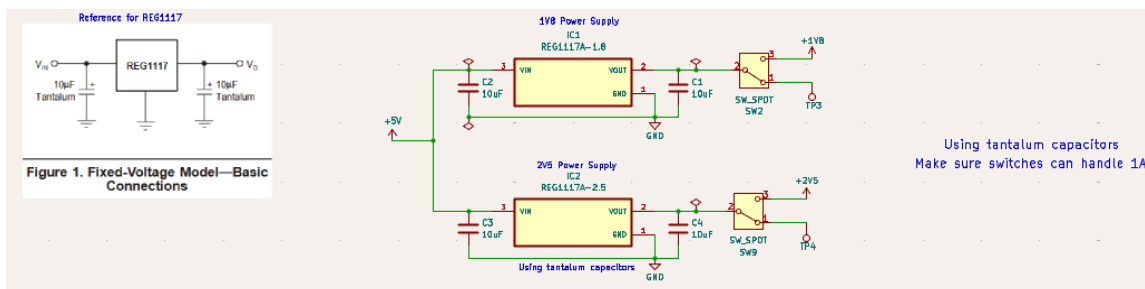


Figure 12. TX board 1.8 V and 2.5 V supplies built from REG1117 regulators

TXB0101 translators shift the SPI (Figure 13) and sequencer (Figure 14) signals from 3.3 V to 1.8 V.

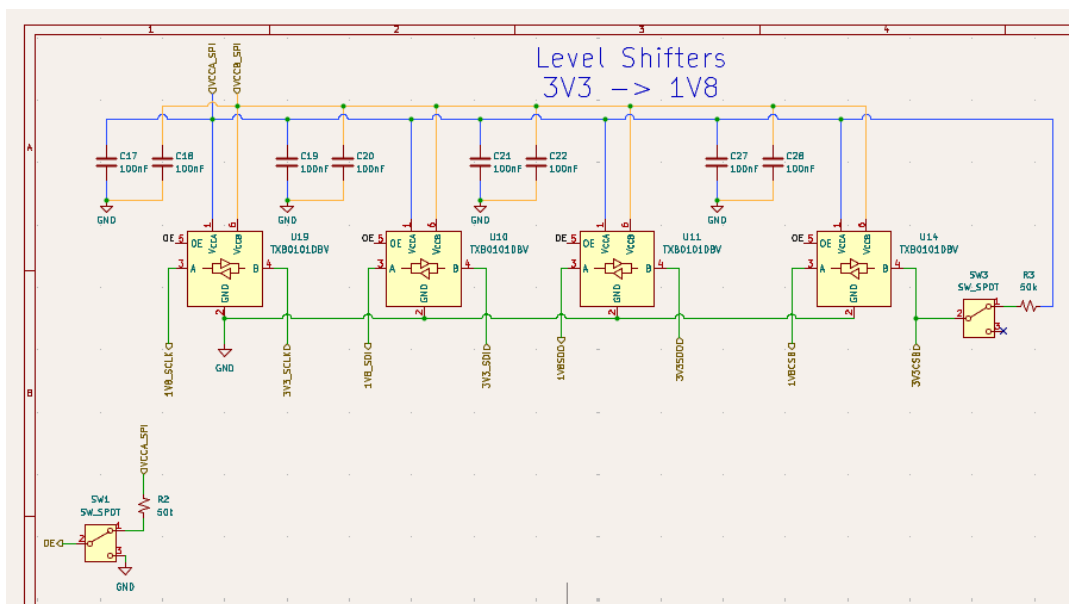


Figure 13. SPI level shifters, 3.3 V to 1.8 V

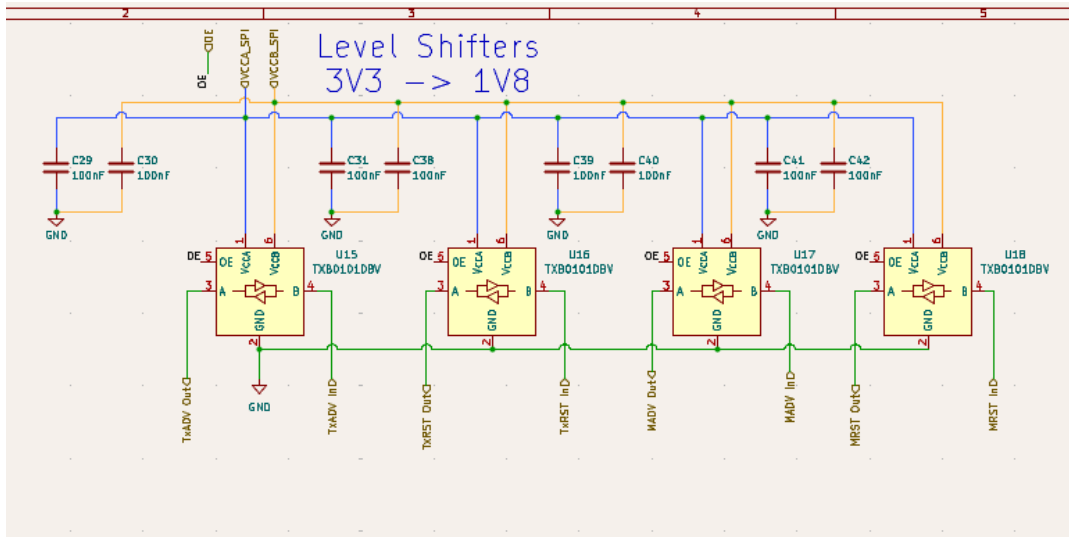


Figure 14. Sequencer level shifters for TxADV, TxRST, MADV and MRST

The eight RF outputs (four differential pairs) must be routed with equal lengths to keep the channels in phase (Figure 15).

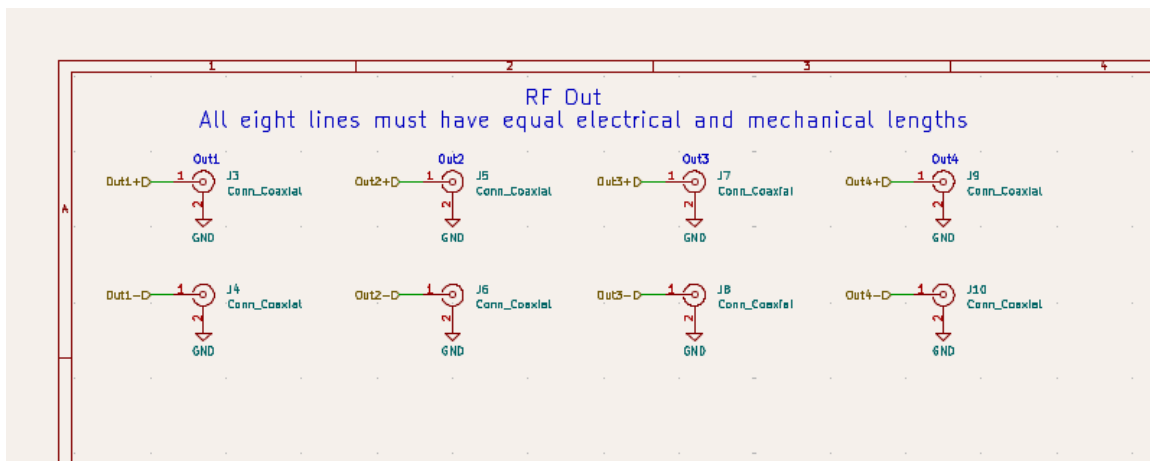


Figure 15. RF output connectors for the four differential output pairs

Figure 16 shows the layout in progress, with parts placed and routing still to be done.

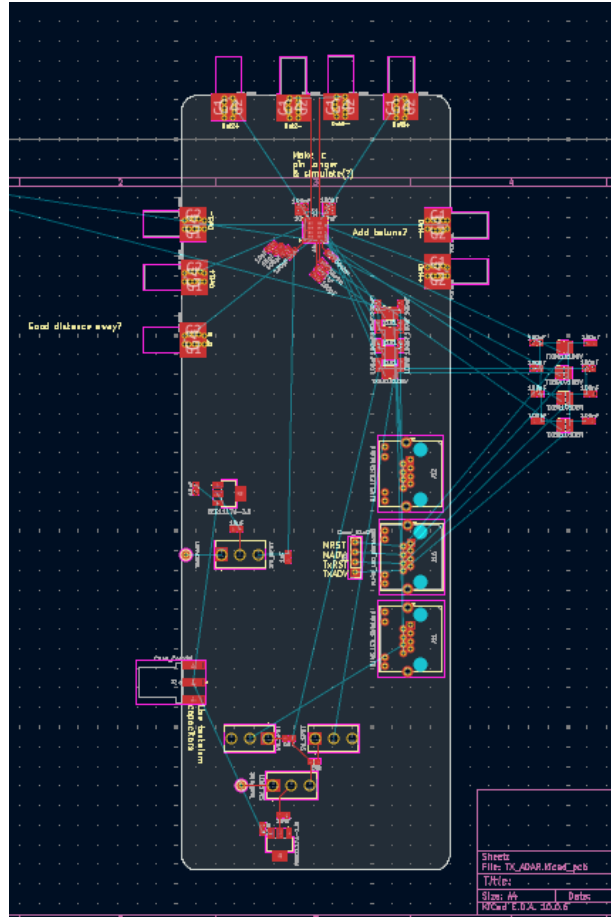


Figure 16. TX up-converter layout in progress

§4 Waveguide and Antenna Design

Both the TX and RX paths need a microstrip-to-waveguide transition. The Vivaldi antenna is now a candidate for that transition.

§4.1 Waveguides and Transitions

Work this week covered the waveguide sections and the transitions that connect the boards to them. The main constraint is the low end of the band.

§4.2 Vivaldi as a Waveguide Transition

The Vivaldi is fed from microstrip through a slotline, and its flared slot launches the signal into the waveguide. The CST model (Figure 17) ends in a waveguide port, so its S11 is the transition match. The goal is S11 below -10 dB from 24 to 40 GHz.

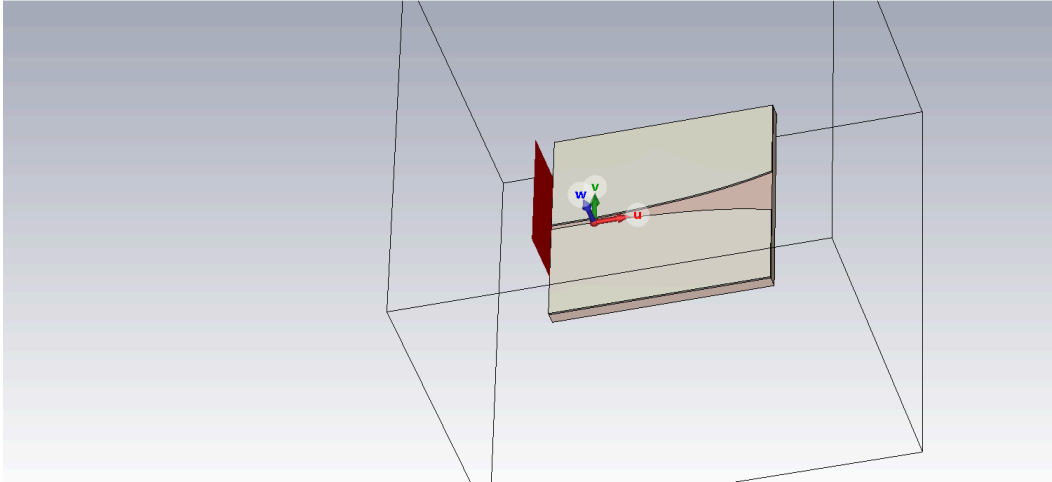


Figure 17. Vivaldi model with its waveguide port in CST

Figure 18 shows the current S11. It is about -6.4 dB at best (near 30 GHz) and about -2 dB at worst (24 GHz). The match improves above 40 GHz, so the structure is currently tuned too high.

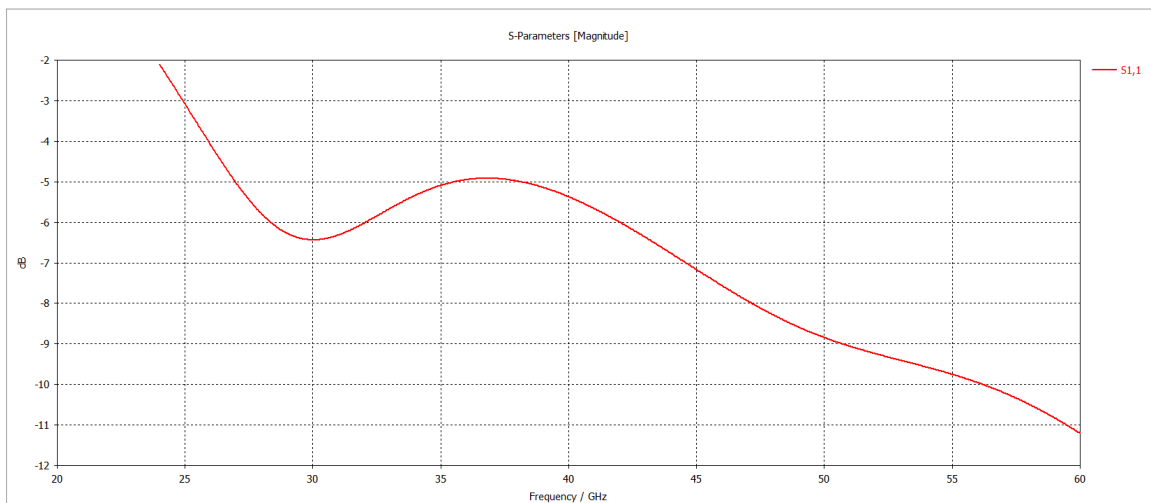


Figure 18. Simulated S11 from 20 to 60 GHz

Widening the S-shaped flare should lower the operating band. The CST optimizer is set to sweep ext, k, r, sL and subw (Figure 19).

	Parameter	Min	Max	Initial	Current	Best
☒	ext	0.065	0.135	0.065	0.065	0
☐	h	0.18	0.22	0.2	0.2	0
☒	k	2.879	5.981	5.981	5.981	0
☐	MW	0.225	0.275	0.25	0.25	0
☐	QWM	0.81	0.99	0.9	0.9	0
☐	QWS	0.675	0.825	0.75	0.75	0
☒	r	0.715	1.485	0.7	0.7	0
☐	s	0.045	0.055	0.05	0.05	0
☒	sL	0.65	1.35	0.65	0.65	0
☒	subw	1.95	4.05	3.6068407403839	3.6068407403839	0
☐	TL	2.7	3.3	3	3	0

Figure 19. CST optimizer parameter list with the five swept parameters checked

§5 Signal Processing

Until the FPGA ADC path is running, the lab spectrum analyzer will sample the IF signal from the receiver board and pass it to a PC (Figure 20). This lets algorithm development start now.

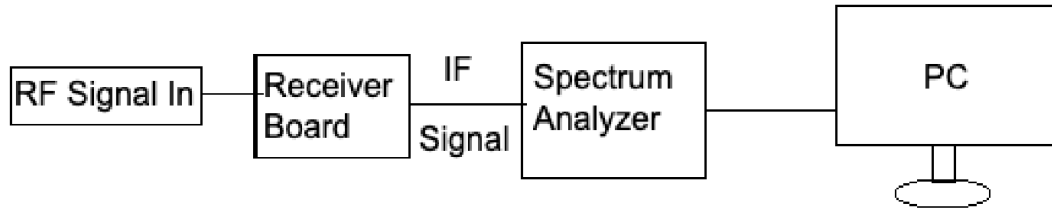


Figure 20. Planned IF capture chain using the lab spectrum analyzer

For a chirp slope S , a target at range R produces a beat frequency:

$$f_b = 2SR / c \quad (3)$$

Target motion adds a Doppler shift of $2v/\lambda$. The algorithms will estimate range and velocity from the sampled IF data.

§6 FPGA Development

The sub-team researched SPI controller architectures for a VHDL implementation, which will be developed on an Opal Kelly board with ADC modules (Figure 21).

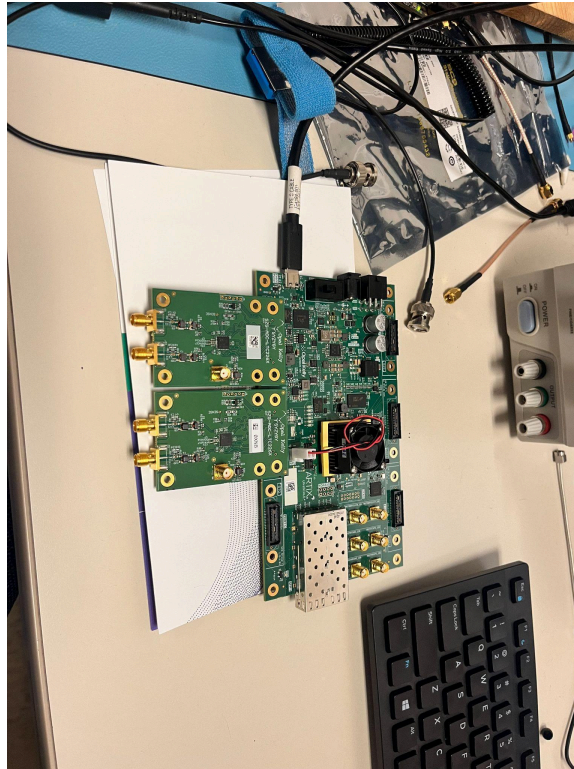


Figure 21. Opal Kelly development board with ADC modules attached

Programming has not started. The options are the Opal Kelly SDK, which the team does not have yet, or JTAG through Vivado. JTAG is preferred because it bypasses the on-board microcontroller.

§7 Research

§7.1 Secondary Research

- **Datasheets:** AD3551R layout guidelines, TXB0101 and REG1117, applied in the schematics above.
- **SPI controller architectures:** reviewed for the VHDL design.
- **PLL research:** for the VCO and synthesizer.
- **Waveguides and transitions:** waveguide cutoff and microstrip-to-waveguide transitions, including tapered-slot and finline types.
- **Vivaldi design:** how the flare geometry sets the lowest usable frequency.

§7.2 Primary Research

No measurements were taken this week. Primary data collection starts once spectrum analyzer sampling is set up and the first boards are assembled.

§8 Pending Issues

- **Waveguide transition:** the Vivaldi is a candidate, not yet final, and must match well at 24 GHz.
- **Vivaldi match:** S11 is not yet below -10 dB across 24 to 40 GHz (best about -6.4 dB, worst about -2 dB).
- **ADAR placement:** still open. It sets the RF trace lengths for the eight equal-length TX outputs.
- **FPGA programming path:** Opal Kelly SDK or JTAG through Vivado.
- **TX up-converter:** the schematic may change, and the layout is not finished.

§9 Individual Contributions

Team Member	Contribution	Weekly Hours	Total Hours
Flatrim Dreshaj	DAC PCB design, waveguide and waveguide transition design, systems design, team organization	5	45
Brett Brunner	Antenna design	7	16
Durotimi Johnson	TX up-converter board	5	20
Ryland Feist	VCO board, PLL research	4	10
Elliott Bourgeois	Signal processing	4	8
Justin Adrowski	FPGA	3	7
Aidan O'Grady	FPGA	4	8
Team Total		32	114

Table 4. Individual contributions and hours

§10 Plans for Next Week

- **Waveguide transitions (Flatrim):** continue the transition design and evaluate the Vivaldi as the transition.

- **Antenna (Brett):** get S11 below -10 dB across 24 to 40 GHz by widening the flare and running the optimizer.
- **Signal processing (Elliott):** set up spectrum analyzer sampling and start DSP algorithm development.
- **TX up-converter (Durotimi):** continue the layout with equal-length RF routing.
- **Ethernet SPI board (Flatrim):** finish the schematic.
- **FPGA (Justin, Aidan):** settle the programming path and start the VHDL SPI controller.
- **VCO (Ryland):** PLL board schematic and research.